instituto superior de engenharia de lisboa

Master Thesis Title, Project Work or Internship Report

JOHN VERY LONGNAME DOE
(Graduated)

Dissertation submitted to obtain the MSc degree in

Electronics and Telecommunications Engineering

SUPERVISORS

PhD Elisabeth Adviser Name Doe
PhD John Adviser Name Doe Male

July of 2026



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July of 2026

Acknowledgments

The acknowledgments. You are free to write this section at your own will. However, usually it starts with the institutional acknowledgments (adviser, institution, grants, workmates, …) and then comes the personal acknowledgments (friends, family, …).

Statement of integrity

I declare that this dissertation is the result of my personal and independent research. Its content is original, and all sources listed in the bibliographic references were consulted and are duly mentioned in the text. I further declare that all scientific and technical references relevant to the development of the work are duly cited and included in the bibliographic references.

The author

Lisbon, July, 21 2026

Master Thesis Title, Project Work or Internship Report

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This document was created using the (pdf)LaTeX processor, based in the "iselthesis" template [5], developed at the DEI of ISEL-UPL.

Master Thesis Title, Project Work or Internship Report

Abstract

This document is both the ISEL LaTeX template for bachelor's project reports and master's dissertations, and its user manual. If you only read this page, read the three rules below:

1. **Compile with LuaLaTeX**, not pdfLaTeX. Only then is the Verdana font required by the ISEL guidelines used.
2. **Edit only** the `Config` folder (options, cover and list of files), the `Chapters` folder (text and images) and the `bibliography.bib` file.
3. **Do not change** `template.tex`, `iselthesis.cls` or the `ISELthesis-files` folder.

Details are in Chapter 2 (structure and options) and Chapter 3 (compiling and examples).

About the abstract. The template places the abstract in the main language of the document first, followed by the abstract in the other language (English or Portuguese). The abstract must not exceed **one page or 300 words**, and should answer four questions: **what is the problem**, **why is it interesting**, **what is the solution**, and **what follows from the solution**.

Keywords: Keywords (in English) …– at least 3 (three)

Master Thesis Title, Project Work or Internship Report

Resumo

Este documento é, ao mesmo tempo, o *template* LaTeX do ISEL para relatórios de projeto de licenciatura e dissertações de mestrado, e o respetivo manual de utilização. Se ler apenas esta página, leia as três regras seguintes:

1. **Compile com LuaLaTeX**, e não com pdfLaTeX. Só assim é usada a fonte Verdana exigida pelas normas do ISEL.
2. **Edite apenas** a pasta `Config` (opções, capa e lista de ficheiros), a pasta `Chapters` (texto e imagens) e o ficheiro `bibliography.bib`.
3. **Não altere** o `template.tex`, o `iselthesis.cls` nem a pasta `ISELthesis-files`.

Os detalhes estão no Capítulo 2 (estrutura e opções) e no Capítulo 3 (compilação e exemplos).

Sobre o resumo. O *template* coloca primeiro o resumo na língua principal do documento e depois o resumo na outra língua (Português ou Inglês). O resumo não deve exceder **uma página nem 300 palavras**, e deve responder a quatro questões: **qual é o problema**, **porque é interessante**, **qual é a solução** e o que **resulta dessa solução**.

Palavras-chave: Palavras-chave (em Português) …– mínimo de 3 (três)

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Acronyms

α	First letter of the greek alphabet 12
β	Second letter of the greek alphabet 12
GCD	Greatest Common Divisor 12
LCM	Least Common Multiple 12

Glossary

matrix	a rectangular table of elements 12
universal set	the set of all things 11

1 INTRODUCTION

This document is the LaTeX template for bachelor's project reports and master's dissertations at the Lisbon School of Engineering (ISEL), Polytechnic University of Lisbon. It follows the ISEL style guide *Normas de escrita e apresentação de trabalhos finais de curso*, and it is also its own user manual: the PDF you are reading was produced by the template, and its source files are examples you can learn from and replace with your own text.

1.1 Before You Start

1. **Compile with LuaLaTeX**, not pdfLaTeX, so that the Verdana font required by ISEL is used (Chapter 3).
2. **Edit only** the `Config` and `Chapters` folders and your bibliography file (Chapter 2).
3. **Do not change** `template.tex`, `iselthesis.cls` or the `ISELthesis-files` folder.

1.2 Organisation of this Document

- **Chapter 2** describes the folders and files of the template, the document options and how to configure your document.
- **Chapter 3** explains how to compile the document and shows examples of common LaTeX constructs: figures, tables, glossaries, algorithms, code listings and citations.

1.3 License and Status

This template is distributed under the MIT License.

This package and template are not official for ISEL/UPL.

Bugs and suggestions are welcome at `https://github.com/matpato/iselthesis`.

2 ISELTHESIS USERS MANUAL

2.1 Introduction

This chapter describes how to use the LaTeX class `iselthesis`. This class is a major rewrite of templates used in several universities, which were in turn adapted from the FCT-UNL template [4]. We aimed at providing an improved visual layout and, simultaneously, a *very easy to use* template (aka, a LaTeX template for dummies).

The main design principle is simple: **you never edit the main file** `template.tex` **nor the class files**. Everything you need to change (document type, language, title, author, chapters, …) is set in the small configuration files inside the `Config` directory.

Text encoding. All your `.tex` and `.bib` files **must** be saved in **UTF-8**. This is the default in all modern editors (Overleaf, TeXstudio, VS Code, TeXShop, …). If accented characters (e.g., "ç", "ã") appear wrong in the PDF, check the encoding of the file in your editor.

2.2 Folder Structure

The template is organised into a main directory (`iselthesis`) with the files and subdirectories listed in Table 2.1. The last column tells you whether you are expected to edit each item.

Table 2.1. Folders and files of the template

Item	Content	Edit?
Your content		
`Chapters/`	Your text files: chapters, abstracts, acknowledgements, statement, glossaries, Gantt diagram, appendices, annexes, etc. Replace the examples with your own.	**yes**
`Chapters/img/`	All the images of your document.	**yes**

Continued on the next page

Table 2.1. Folders and files of the template (Cont.)

Item	Content	Edit?
`Chapters/scripts/`	Code or pseudo-code to be included in the text.	**yes**
`bibliography.bib`	Your bibliographic references (see below).	**yes**
Configuration (folder `Config/`)		
`/_options.tex`	Document options: degree, type, language, media, etc. (Section 2.3).	**yes**
`/_cover.tex`	Cover contents: title, author, advisers, committee, date.	**yes**
`/_files.tex`	Which files are included: chapters, abstracts, appendices, etc.	**yes**
`/_memoir.tex`	Paper size, font size and `draft`/`final` mode.	**yes**
`/_packages.tex`	Your own packages and commands.	**yes**
`/_institution.tex`	University and school names.	only if not ISEL
`template.xmpdata`	PDF/A metadata: author, title, keywords, language. Must match `Config/_cover.tex`.	**yes**
Template (do not edit)		
`template.tex`	The main file: run LuaLaTeX on this one.	**no**
`iselthesis.cls`	The class file.	**no**
`ISELthesis-files/`	Internal files of the class. **None of them should be changed**:	**no**
`Chap-Styles/`	Chapter heading styles.	**no**
`Default/`	Institutional defaults: cover layout, margins, colours, section styles (`defaults.clo`) and course definitions (`courses.clo`).	**no**
`Font-Styles/`	Font configuration and the Verdana font files.	**no**
`Strings/`	Fixed texts of the template in English and Portuguese (cover, abstract, statement, etc.).	**no**
`user-sty/`	Auxiliary packages used by the class.	**no**
`fix-babel.clo`	Corrections to the `babel` translations.	**no**
`lang-conf.clo`	Configuration of the supported languages.	**no**
`manual-bib.bib`	References cited in this manual.	**no**
`options.clo`	Definition of the document options and their defaults.	**no**
`packages.clo`	Packages loaded by the class.	**no**
`version.sty`	Version of the template.	**no**

Continued on the next page

Table 2.1. Folders and files of the template (Cont.)

Item	Content	Edit?
`Logo/`	Institutional logos and cover backgrounds.	**no**
`Scripts/`	Utility scripts, e.g., to clean temporary files.	**no**

A few notes:

- **Excluding a file.** To leave a file out of the document, remove it from `Config/_files.tex`, or rename it (e.g., `ganttdiagram_.tex` instead of `ganttdiagram.tex`).
- **Statement.** The statement file is required for the final version of the master's dissertation (`docdegree=msc`). It exists in English and Portuguese; the one matching the main language of the document is used.
- **References.** An easy way to obtain BibTeX entries is to enable `Show links to import citation into BibTeX` in the Google Scholar settings.

Again, we would like to recall that all your LaTeX files should be stored in the `Chapters` directory, and all the images in the `Chapters/img` directory.

2.3 Document Options

The document options are **not** given in the `\documentclass[…]` line of `template.tex`. Instead, they are set in the file `Config/_options.tex`, using the command `\iselthesissetup`:

```
\iselthesissetup{
  docdegree      = msc,          % msc, mscprep, bsc
  doctype        = dissertation, % dissertation, project
  lang           = en,           % en, pt
  coverlang      = en,           % en, pt
  printcommittee = true,         % true, false
  media          = screen,       % screen, paper
  linkscolor     = darkblue,     % darkblue, black
}
```

Any option omitted from this file keeps its default value. Options are applied in the following order, and the last one wins:

1. the institutional defaults (in `ISELthesis-files/options.clo`);
2. options given in `\documentclass[…]`, if any (kept only for compatibility with older versions of the template);
3. the options in `Config/_options.tex`.

Some options are fixed by the institution, to guarantee that all documents follow the ISEL graphic guidelines, and should not be changed: the font (Verdana, or a similar sans-serif font), the chapter style (`isel`), the place of the lists of figures and tables (`front`) and the bibliography style (numeric, `biblatex`).

Table 2.2 lists the available options. For each option, the default value is marked with an asterisk (*).

Table 2.2. Description of the document options

Option	Definition and values	Set in
docdegree	**Document degree**: master's dissertation – msc(*), preparation of the master's dissertation – mscprep, and bachelor's report – bsc.	`_options.tex`
doctype	**Type of document**: dissertation(*) or project.	`_options.tex`
lang	**Main language of the document**: English – en(*) or Portuguese – pt.	`_options.tex`
coverlang	**Language of the cover pages**: English – en or Portuguese – pt(*).	`_options.tex`
printcommittee	**Print the list of committee members** on the second cover: true(*) or false. Set to false in submitted versions, before the committee is appointed.	`_options.tex`
media	**Target of the PDF**: screen(*) or paper (see Section 2.3.3).	`_options.tex`
linkscolor	**Colour of the hyperlinks in the PDF**: dark-blue(*) or black (set to black for printing).	`_options.tex`
memoir	**Options of the base class** `memoir`: a4paper, 11pt, draft or final. Set with `\ncmemoirsetup{…}`.	`_memoir.tex`
fontstyle	**Font of the document**: Verdana (fixed by the institution).	fixed
chapstyle	**Chapter style**: isel (fixed by the institution).	fixed
otherlistsat	**Place of the other lists** (figures, tables, …): front (fixed by the institution).	fixed
biblatex	**Options of** `biblatex`: numeric style, sorted by name/year/title, with back references (fixed by the institution).	fixed

2.3.1 Language Related Options

You must choose the main language of the document (option `lang`). The available values are:

1. **en** — The document is written in English (with an abstract in Portuguese).
2. **pt** — The document is written in Portuguese (with an abstract in English).

The language option affects:

- **The order of the abstracts.** First the abstract in the main language and then the one in the other language. If the main language is English, you will see first the "Abstract" and then the "Resumo". If you switch the main language, the order is switched automatically.
- **The names of the document parts.** E.g., "Chapter" vs. "Capítulo", "Contents" vs. "Índice", "Figure" vs. "Figura", etc.
- **The labels in the bibliography.** E.g., "Technical Report" vs. "Relatório Técnico", "MSc Thesis" vs. "Tese de Mestrado", etc.

The option `coverlang` sets the language of the cover pages independently of the main language. In most cases it should be the same as `lang`.

No matter which language you choose, you will always have the appropriate hyphenation rules for the language in use at each point: Portuguese rules in the "Resumo", English rules in the "Abstract", and the main language rules in the rest of the document. If you need to force the hyphenation of a word, write it inside `\hyphenation{}` in `Config/_packages.tex`, e.g., `\hyphenation{op-ti-cal net-works}`.

2.3.2 Class of Text

You must choose the class of text for the document (option `docdegree`). The available values are:

1. **bsc** — Bachelor's (BSc) project report. The second cover page includes signature lines for the author and the advisers.
2. **mscprep** — Preparation of the MSc dissertation. This is a preliminary report that ISEL/IPL master's students prepare to conclude the first semester of the two-semester MSc work. The files specified by a) `\dedicatoryfile`, b) `\acknowledgementsfile` and c) `\quotefile` are ignored, even if present, for this class of document. There is no second cover page.
3. **msc** — MSc dissertation. Includes a second cover page with the advisers and the committee, and the statement and copyright pages.

2.3.3 Printing

You must choose the target media of your document (option `media`). The available values are:

1. **screen**(*) — For reading on screen: left and right margins are equal and hyperlinks are coloured;
2. **paper** — For printing: left and right margins are different (wider binding margin) and hyperlinks are black.

Article 50 of Decree-Law No. 115/2013 requires the deposit of a digital copy of doctoral theses and master's dissertations in a repository that is part of RCAAP[1], https://www.rcaap.pt. This deposit aims to preserve scientific work and to provide Open Access to scientific production that is not subject to restrictions or embargo.

For this reason, we include the option to format your document so that it presents well on screen and/or on paper. But always remember that your work will be stored in the RCAAP portal in electronic format.

2.3.4 Draft and Final Versions

The options of the base class `memoir` are set in `Config/_memoir.tex`:

```
\ncmemoirsetup{a4paper,11pt,final}   % draft | final
```

While writing, you may use `draft`: the cover shows "DRAFT" and the current date. Replace it with `final` for the version you submit.

2.3.5 Examples

Let's have a look at a few examples of `Config/_options.tex`:

- BSc project report, in Portuguese, to be read on screen:

  ```
  \iselthesissetup{docdegree=bsc, lang=pt, coverlang=pt}
  ```

- Preparation of the MSc dissertation, in Portuguese, with black links:

  ```
  \iselthesissetup{docdegree=mscprep, lang=pt, coverlang=pt,
                   linkscolor=black}
  ```

- MSc project, in English, to be printed on paper:

  ```
  \iselthesissetup{docdegree=msc, doctype=project, lang=en,
                   coverlang=en, media=paper, linkscolor=black}
  ```

The present document uses the following `Config/_options.tex`:

[1] Repositórios Científicos de Acesso Aberto de Portugal

```
\iselthesissetup{
  docdegree      = msc,
  doctype        = dissertation,
  lang           = pt,
  coverlang      = pt,
  printcommittee = true,
  media          = screen,
  linkscolor     = darkblue,
}
```

2.4 How to Write Using LaTeX

Please have a look at Chapter 3, where you may find many examples of LaTeX constructs, such as sectioning, inserting figures and tables, writing equations, theorems and algorithms, code listings, etc.

3 A SHORT LaTeX TUTORIAL WITH EXAMPLES

This Chapter aims at exemplifying how to do common stuff with LaTeX. We also show some stuff which is not that common! ;)

Please, use these examples as a starting point, but you should always consider using the *Big Oracle* (aka, Google, your best friend) to search for additional information or alternative ways for achieving similar results.

IMPORTANT — Compile with LuaLaTeX.

The ISEL graphic guidelines require the **Verdana** font. Verdana is a system (TrueType/OpenType) font, which can only be used by **LuaLaTeX** (or XeLaTeX). Therefore, this template **must be compiled with LuaLaTeX**.

- If you compile with `pdflatex`, the document is still produced, but **without Verdana**: a Helvetica-like font is used instead, and the document does **not** follow the ISEL guidelines.
- The Verdana font files are included in the template. There is nothing to install.
- See Section 3.5 to configure your editor.

3.1 Document Structure

In engineering and science, a thesis or dissertation is the culmination of a master's or Ph.D. degree. A thesis or dissertation presents the research that the student performed for that degree. From the student's perspective, the primary purpose of a thesis or dissertation is to persuade the student's committee that he or she has performed and communicated research worthy of the degree. In other words, the main purpose of the thesis or dissertation is to help the student secure the degree.

From the perspective of the engineering and scientific community, the primary purpose is to document the student's research. Although much research from theses and dissertations is also communicated in journal articles, theses and dissertations stand as detailed documents that allow others to see what the work was and how it was performed. For that reason, theses and dissertations are often read by other

graduate students, especially those working in the research group of the authoring student [1, 2, 6, 7].

A thesis or dissertation is expected to have the following structure, organised in three parts (optional elements are marked with *):

Front matter	Main text	Back matter
Cover	Introduction	References
Acknowledgements*	State of the art / Related work	Gantt diagram*
Abstracts (PT and EN)	Proposed method	Appendices*
Table of Contents	Experimental results	Annexes*
Lists of figures, tables, listings and algorithms	Conclusions and future work	
Acronyms, symbols and glossary*		

The front and back matter are generated automatically by the template from the files listed in `Config/_files.tex`; you only need to write the content.

3.1.1 State-of-the-Art

State-of-the-Art (SoTA) is a step to demonstrate the novelty of your research results. The importance of being the first to demonstrate research results is a cornerstone of the research business[1].

Besides demonstrating the novelty of your research results, a SoTA has other important properties:

1. It teaches you a lot about your research problem. By reading literature related to your research problem you will learn from other researchers and it will be easier for you to understand and analyse your problem;
2. It proves that your research problem has relevance;
3. It shows different approaches to a solution;
4. It shows what you can reuse from what others have done.

3.1.2 Related work

In the *Related Works* section, you should discuss briefly about published matter that technically relates to your proposed work[2].

A short summary of what you can include (but not limited to) in the Related Works section:

1. Work that proposes a different method to solve the same problem;

[1] "Why and how to write the state-of-the-art", by Babak A. Farshchian, May 22, 2007

[2] https://academia.stackexchange.com/questions/68164/how-to-write-a-related-work-section-in-computer-science

2. Work that uses the same proposed method to solve a different problem;
3. A method that is similar to your method that solves a relatively similar problem;
4. A discussion of a set of related problems that covers your problem domain.

3.2 Glossary and Nomenclature/List of Symbols

Many technical documents use terms or acronyms unknown to the general population. It is common practice to add a glossary to make such documents more accessible. A *glossary* is a nice thing to have in a report and usually very helpful. As you probably can imaging, it is very easy to create in LaTeX.

As with all packages, you need to load glossaries with `\usepackage`, but there are certain packages that must be loaded before glossaries, if they are required: `hyperref`, `babel`, `polyglossia`, `inputenc` and `fontenc`.

```
\usepackage{glossaries}
\makenoidxglossaries
```

Once you have loaded `glossaries`, you need to define your terms in the preamble (or, separated file) and then you can use them throughout the document.

Next you need to define the terms you want to appear in the glossary. Again, this must be done in the preamble. This is done using the command

`\newglossaryentry{<label>}{<key-val list>}`

The first argument `<label>` is a unique label to allow you to refer to this entry in your document text. The entry will only appear in the glossary if you have referred to it in the document using one of the commands listed later. The second argument is a comma separated `<key>=<value>` list.

Inside the text you just need to use the command `\gls{name}` or `\glspl{name}` (plural name) to call it. For example, the following defines the term 'set' and assigns a brief description. The term is given the label set. This is the minimum amount of information you must give:

```
\newglossaryentry{set} % the label
{ name=set,                    % the term
  description={a collection of objects}  % a brief description
}
```

Other example, now the glossary associated with a symbol, universal set:

```
\newglossaryentry{U}   % the label
{ name={universal set}, % the term
  description={the set of all things}   % a brief description
  symbol={\ensuremath{\mathcal{U}}}   % the associated symbol
```

```
}
```

Here's a simple example:

```
\usepackage{glossaries}
\newglossaryentry{ex}{name={sample},description={an example}}
\newacronym{svm}{SVM}{support vector machine}
\newacronym{beta}{$\beta$}{Second letter of the greek alphabet}
\newacronym{alpha}{$\alpha$}{First letter of the greek alphabet}

\begin{document}
Here's my \gls{ex} term. First use: \gls{svm}.
Second use: \gls{svm}.

\textit{I want the \gls{beta} to be listed after the \gls{alpha}}.
\end{document}
```

This produces: *Here's my sample term. First use: support vector machine (SVM). Second use: SVM.*
I want the Second letter of the greek alphabet (β) to be listed after the First letter of the greek alphabet (α).

Do not use `\gls` in chapter or section headings as it can have some unpleasant side-effects. Instead use `\glsentrytext` for regular entries and one of `\glsentryshort`, `\glsentrylong` or `\glsentryfull` for acronyms. Alternatively use glossaries-extra which provides special commands for use in section headings, such as `\glsfmtshort{<label>}`.

The plural of the word "matrix" is "matrices" not "matrixs", so the term needs the plural form set explicitly:

```
\newglossaryentry{matrix}% the label
{ name=matrix, % the term
  description={a rectangular table of elements},
 plural=matrices % the plural
}
```

Given a set of numbers, there are elementary methods to compute its Greatest Common Divisor, which is abbreviated GCD. This process is similar to that used for the Least Common Multiple (LCM).

3.3 Importing Images

You can import external graphics using package `graphicx` (already loaded by the template). You will find several examples here: https://www.overleaf.com/learn/lat

ex/Inserting_Images. The most important command is \includegraphics. LaTeX itself treats the image like normal text, i.e. as a box of certain height and width. The package documentation list the options width and height, as well as others.

This syntax is fairly self-explanatory, however, one thing to note is the use of the \centering command. There are other ways to center a figure, but this is the correct way to do it inside a float environment. You will almost always be using this command when using floats.

```
\begin{figure}[H]
\centering
\includegraphics[scale=1.5]{atom.png}
\end{figure}
```

3.4 Floats Figures and Tables, and Captions

The tabular environment can be used to typeset tables with optional horizontal and vertical lines. LaTeX determines the width of the columns automatically. The first line of the environment has the form: \begin{tabular}[pos]{table spec}, where (1) table spec tells LaTeX the alignment to be used in each column and the vertical lines to insert; and, (2) pos can be used to specify the vertical position of the table relative to the baseline of the surrounding text. The number of columns does not need to be specified as it is inferred by looking at the number of arguments provided. It is also possible to add vertical lines between the columns here. You can draw beautiful tables with the booktabs package https://ctan.org/pkg/booktabs?lang=en.

Some notes are important to followed, such as present in Table 3.1:

i) Not defined vertical lines;

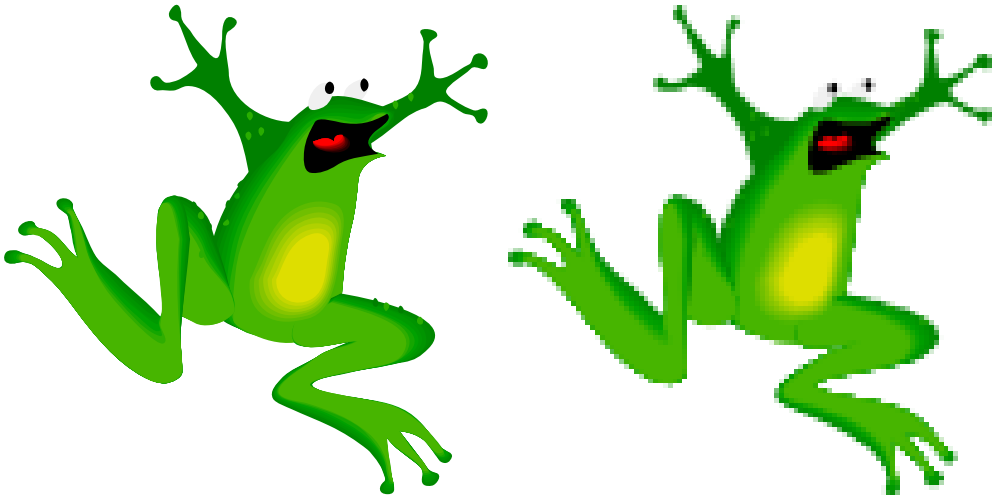
ii) The legend must be on top;

iii) Use \toprule, \midrule and \bottomrule to draw horizontal lines.

Table 3.1. Table's rules.

Item		
Animal	Description	Price ($)
Gnat	per gram	13.65
	each	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

Since the template is compiled with LuaLaTeX (see Section 3.5), images must be in **PDF, PNG or JPEG** format. PostScript images (EPS/PS) are not supported directly; convert them to PDF first (e.g., with epstopdf).



(a) *Vectorial* **(b)** *Raster*

Figure 3.1. *Subfigure example with vectorial and no-vectorial images*

To get the best quality of the images in PDF files I'd recommend using vector-based graphics for images. The best format to save images in is .pdf, see Figure 3.1a. With programs like `Inkscape`, you can draw as you would in MS Paint (and do much more), and because the images are vector-based instead of pixel-based, their quality should be preserved when converting to PDF in any way.

In all cases, each image must be in an individual 1-image file; no animation files or multipage documents.

There are two different ways to place two figures/tables side-by-side. More complicated figures with multiple images. You can do this using subcaption environments[3] inside a figure environment. Subcaption will alphabetically number your figures and you have access to the complete reference as usual through `\ref{fig:figurelabel}`, Figure 3.1, or Figure 3.1b using `\ref{fig:subfigurelabel}`.

Using the package listings you can add non-formatted text as you would do with `\begin{verbatim}` but its main aim is to include the source code of any programming language within your document. If you wish to include pseudo-code or algorithms see LaTeX/Algorithms_ and_Pseudocode, as Algorithm 3.1.

```
1  # comentário
2  square <- function(x) {
3      x^2
4      % |$x^{2}$|
5  }
6
7  # nerv
8  x <- c(1:100)
9  y <- square(x)
```

Listing 3.1. R-Code (Test).

[3]The subcaption package: http://mirrors.ibiblio.org/CTAN/macros/latex/contrib/caption/subcaption.pdf

Algorithm 3.1 An algorithm with caption

Require: $n \geq 0$
Ensure: $y = x^n$
  $y \leftarrow 1$
  $X \leftarrow x$
  $N \leftarrow n$
  while $N \neq 0$ **do**
    if N is even **then**
      $X \leftarrow X \times X$
      $N \leftarrow \frac{N}{2}$ ▷ This is a comment
    else if N is odd **then**
      $y \leftarrow y \times X$
      $N \leftarrow N - 1$
    end if
  end while

3.5 Generating PDFs from LaTeX

3.5.1 Compiling with LuaLaTeX

This template must be compiled with LuaLaTeX. Only LuaLaTeX (or XeLaTeX) can use the Verdana font required by the ISEL guidelines. With `pdflatex` the document compiles, but with the wrong font.

A typical pass for a document with figures, cross-references and a bibliography is:

```
$ lualatex template
$ bibtex   template
$ lualatex template
$ lualatex template
```

To be sure that the table of contents, cross-references and bibliography citations are up-to-date, you must run `lualatex` once, then `bibtex`, and then `lualatex` twice. The simplest way is to let `latexmk` do all the passes for you:

```
$ latexmk -lualatex template
```

You will notice that there is a new PDF file in the working directory called `template.pdf`. Simple :)

LuaLaTeX is slower than `pdflatex`, especially in the first run, when it builds its font cache. This is normal.

3.5.2 Configuring your Editor

Most editors use `pdflatex` by default, so you must select LuaLaTeX explicitly:

- **Overleaf**: *Menu* → *Settings* → *Compiler* → `LuaLaTeX`.

- **TeXstudio**: *Options* → *Configure TeXstudio* → *Build* → *Default Compiler* → `LuaLaTeX`.
- **VS Code** (LaTeX Workshop extension): choose the recipe `latexmk (lualatex)`.
- **TeXShop**: *Typeset* → select `LuaLaTeX` in the toolbar menu.

You can also force the compiler from the source file. Some editors (TeXstudio, TeXShop, VS Code) read a "magic comment" in the first line of `template.tex`:

```
% !TEX program = lualatex
```

3.5.3 Checking the Font

To confirm that Verdana was used, open the PDF and check the document properties (*File* → *Properties* → *Fonts*). Verdana must be listed. If you see Helvetica, TeX Gyre Heros or Nimbus Sans instead, the document was compiled with `pdflatex`.

If LuaLaTeX stops with an error saying that the font `Verdana` cannot be found, check that the folder `ISELthesis-files/Font-Styles` was copied together with the rest of the template (e.g., when uploading it to Overleaf).

3.5.4 Dealing with Images

LuaLaTeX accepts images in PDF, PNG and JPEG format. I strongly recommend you to use PDF figures in vectorial format (do not use bitmap images unless you have no other choice).

Do not include the extension of the figure file in the `\includegraphics` command, for instance use `\includegraphics{filename}` and not `\includegraphics{filename.png}`. LuaLaTeX will find the appropriate file, preferring the PDF version if several exist.

3.6 Bibliography and Footnotes

If you are writing only one or two documents and aren't planning on writing more on the same subject for a long time, maybe you don't want to waste time creating a database of references you are never going to use. In this case you should consider using the basic and simple bibliography support that is embedded within LaTeX.

LaTeX provides an environment called `thebibliography` that you have to use where you want the bibliography; that usually means at the very end of your document, just before the `\end{document}` command. Here is a practical example:

```
\begin{thebibliography}{9}
\bibitem{lamport94}
  Leslie Lamport,
  \emph{\LaTeX: A Document Preparation System}.
  Addison Wesley, Massachusetts,
```

```
  2nd Edition,
  1994.
\end{thebibliography}
```

In this document, the bibliography is in a separate document: `bibliography.bib` where information is entered from `https://scholar.google.pt/`, as show Figure 3.2.

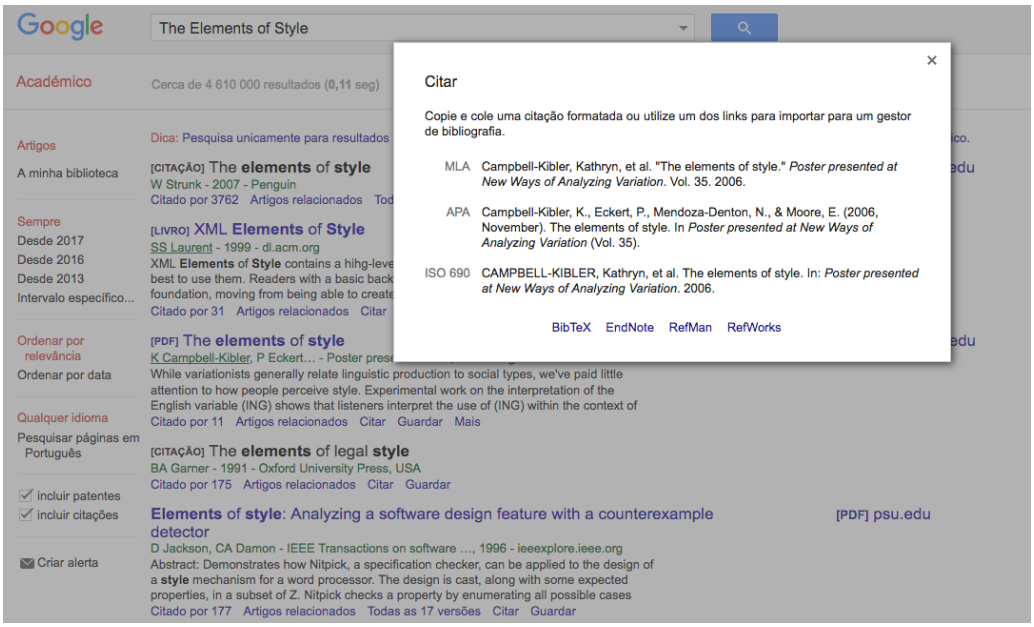


Figure 3.2. *Screenshot from Scholar Google*

To actually, cite a given document is *very* easy. Go to the point where you want the citation to appear, and use the following: `\cite{citekey}`, where the `citekey` is that of the `bibitem` you wish to cite, e.g `~\cite{lamport94}`. When LaTeX processes the document, the citation will be cross-referenced with the bibitems and replaced with the appropriate number citation. The advantage here, once again, is that LaTeX looks after the numbering for you.

When a sequence of multiple citations are needed, you should use a single `\cite{}` command. The citations are then separated by commas. Note that you must not use spaces between the citations. Here's an result example [3, 6, 7].

Footnotes are a very useful way of providing extra information to the reader. Usually, it is non-essential information which can be placed at the bottom of the page. This keeps the main body of text concise.

The footnote facility is easy to use: `\footnote{Simple footnote}`[4].

3.7 Equations

Typesetting mathematics is one of LaTeX's greatest strengths. It is also a large topic due to the existence of so much mathematical notation. It is recommend to read

[4] Simple footnote

the Short Math Guide for LaTeX of the American Mathematical Society.

3.8 Page orientation

The default page layout is "portrait", but sometimes it is still useful/necessary to have the whole document or only single pages changed to "landscape". The latter might be due to a large table or figure. If you want to make appear the left side up, better readable on screen, the pdflscape-package will do it: `\usepackage{pdflscape}`
and again:

```
\begin{landscape}
...
\end{landscape}
```

or, `\includepdf[landscape=true,pages={1}]{example.pdf}`
to put the page in "landscape", while the rest will remain in "portrait" orientation. Nevertheless, the header/footer will also be changed in orientation.

Written by Matilde Pós-de-Mina Pato with collaboration of Nuno Datia, 2012 October (1st version)

4 LOREM IPSUM

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[5] M. Pato. *The ISELthesis LaTeX Template's Manual*. Instituto Superior de Engenharia de Lisboa (ISEL-IPL). 2024. URL: https://github.com/matpato/iselthesis (cit. on p. iii).

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[7] G. Webster. *The Chicago Manual of Style*. 17th edition. University of Chicago Press, 2017, pp. 400–401. ISBN: 978-0-226-28705-8 (cit. on pp. 10, 17).

A APPLIED SURVIVAL ANALYSIS BY HOSMER AND LEMESHOW

Stata Textbook Examples Applied Survival Analysis by Hosmer and Lemeshow

The data files used for the examples in this text can be downloaded in a zip file from the Wiley FTP website or the Stata Web site.

```
# The R package(s) needed for this chapter is the survival package.
# We currently use R 2.0.1 patched version. You may want to make sure
# that packages on your local machine are up to date. You can perform
# updating in R using update.packages() function.

# url: http://www.ats.ucla.edu/stat/r/examples/
# data set is hmohiv.csv.
hmohiv<-read.table("http://www.ats.ucla.edu/stat/r/examples/asa/hmohiv.csv", sep="," , header = TRUE)
attach(hmohiv)
hmohiv

# using the hmohiv data set. To control the type of symbol, a variable called psymbol is created.
# It takes value 1 and 2, so the symbol type will be 1 and 2.
psymbol<-censor+1
table(psymbol)

plot(age, time, pch=(psymbol))
legend(40, 60, c("Censor=1", "Censor=0"), pch=(psymbol))

age1<-1000/age
plot(age1, time, pch=(psymbol))
legend(40, 30, c("Censor=1", "Censor=0"), pch=(psymbol))

# Package "survival" is needed for this analysis and for most of the analyses in the book.
library(survival)
test <- survreg( Surv(time, censor) ~ age, dist="exponential")
summary(test)

pred <- predict(test, type="response")
ord<-order(age)
age_ord<-age[ord]
pred_ord<-pred[ord]
plot(age, time, pch=(psymbol))
lines(age_ord, pred_ord)
legend(40, 60, c("Censor=1", "Censor=0"), pch=(psymbol))
```

I ANNEX 1 LOREM IPSUM

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.